

Minseok Park

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Education

Hanyang University | Seoul, Republic of Korea

B. S. Electronic Engineering | Expected Feb 2027

- GPA: 4.21/4.5

University of North Dakota | Grand Forks, ND

Exchange Program, Electrical Engineering | Jan - May 2026

- GPA: 4.0/4.0 | Advisor: Prof. Jueming Hu

Publication

Park, M., Yang, C. X., Hu, J. (2026). Robust fault diagnosis of rotating machinery using entropy-augmented CNN in time and frequency domains.

- Accepted for the ASME International Mechanical Engineering Congress and Exposition (IMECE).

Research Experience

Resilient UAV Navigation under Sensor Dropout

University of North Dakota | Jan 2026 - Present

- Developed a UAV navigation policy resilient to sensor dropout, achieving up to 91–95% task completion rate and outperforming baseline models under intermittent sensor failure conditions. Manuscript in preparation, targeting IEEE RA-L.

Robust Fault Diagnosis of Rotating Machinery via Entropy-Augmented CNN

University of North Dakota | Jan 2026 - May 2026

- Developed a fault detection framework using entropy-based signal processing and CNN classification to identify mechanical imbalance and misalignment, achieving over 95% classification accuracy across varying rotational speeds. Accepted at ASME IMECE 2026.

End-to-End RL-Based UAV Waypoint Navigation

University of North Dakota | Jan 2026 - May 2026

- Developed an end-to-end RL-based navigation controller simulating Crazyflie UAV dynamics in PyBullet, implementing single and multi-waypoint goal tracking.

Multi-Anchor CNN with ToA Attention for UWB Indoor Localization

Hanyang University | Sep 2025 - Dec 2025

- Developed a Multi-Anchor CNN model integrating ToA-based attention to estimate receiver position for UWB-based indoor localization.

Technical Skills

Programming Languages: Python

AI & Machine Learning: PyTorch, CNN, Reinforcement Learning (RL), Imitation Learning (IL)

Robotics & Simulation: PyBullet, OpenAI Gym, Crazyflie simulation